29 October 2026

The Editors
PLOS ONE

Re: Research Article submission, "Does PageRank find the smart contracts users rely on sooner than counting? A crawl-ordering test on an Ethereum rollup"
Running title: *Crawl-ordering test of PageRank on a rollup's smart contracts*

Dear Editors,

We submit the above manuscript for consideration as a Research Article in *PLOS ONE*. An Ethereum rollup holds millions of smart contracts, and wallets, block explorers, auditors and integrators must choose which of them to examine first. On the web, a crawler that ordered its visits by PageRank computed on the pages read so far reached important pages sooner than one that counted backlinks (Cho, Garcia-Molina and Page, 1998). The manuscript repeats that crawl-ordering experiment on the contracts of the Arbitrum One rollup. Its graph holds contracts only, joined by token transfers and by allowances counted up to the balance of the authorizing contract, and it measures importance on the demand of user accounts, which no ordering reads.

Why the work matters beyond its case study. Graph ranking is often proposed for finding important contracts and is usually checked against statistics of its own edges or against other walks. Our test checks walks against plain counts and against an importance they do not read. In the run that decides the test, a crawl ordered by a walk that couples the two edge types reached the contracts with the most user-account demand sooner than a breadth-first crawl but not sooner than a crawl ordered by the number of contract backlinks, which was sooner. One walk, the published transfer-based AWP, did exceed the count it smooths; walks on allowance edges alone did not on inflow; and where a visit reveals out-edges only the coupled walk led both comparators, in a sensitivity analysis that carries no verdict. Every crawl stayed far from the ideal order. With scores frozen before two three-month windows, PageRank on allowances ranked later approvals below the count of approving contracts, and AWP ranked later senders slightly above the transfer count overall and, in the first window, below it among contracts that were already paid by others.

Why *PLOS ONE*. The journal assesses scientific soundness rather than perceived impact and publishes null and negative results. Our lead result is negative, the coupled walk did not order a crawl better than a count of backlinks, and several others are losses of a walk to a count. We also report openly what was fixed when: an analysis plan and a registration fixed the registered analysis, in raw token units, before its data were extracted; after its results were known we defined the analysis we report, on a graph of contracts with amounts in US dollars over five tokens, scores from the twelve months before each freeze and the balance cap, together with the crawl-ordering test. No result of this analysis is presented as a registered one, the test decides no rule of the plan or the registration, the registered results are set beside the main ones, and excerpts of the plan and the registration are given with the commits and hashes of the full files.

Statements. The manuscript is not under consideration elsewhere and has not been published. It is derived from the first author's doctoral research at the University of South Africa (UNISA). A related manuscript by the same authors, on the coupled walk itself, the decision rules fixed for it and the Sybil exposure of the allowance layer, is being submitted at the same time to *IEEE Access*; we upload it as a related manuscript file. The two papers share the data, the account types and the graphs, and both report the spender holdout coefficients of the raw counts, EndorseRank and AWP with their count contrasts, the closed form of EndorseRank and the solver runtimes. The crawl-ordering test is reported only in this manuscript; the rules for the coupled walk, its weight sweep, its seeded ablation and the Sybil calibration are reported only in the related one. The code, configuration, queries, result summaries, per-contract scores and labels, and the documents that define the analyses are publicly available at `https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation` and `https://github.com/abitocodes/phd-dissertation-draft-en`. The study uses only public blockchain data and public token prices and involves no human participants; only contracts are scored. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors received no specific funding for this work. Claude (Anthropic) was used to write and run analysis code, generate tables and figures, and draft and revise the text; the full statement is in Methods (Use of generative AI).

Thank you for considering our work.

Yours sincerely,

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